Animal Kingdom – Set 2

(Protozoa to Mammalia)

65 HARD-level MCQs (NEW & unique from Set 1) covering all 19 topics of the syllabus
With answer key & solutions

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Basis of Classification & Taxonomic Principles

Question 1

HARD

Which of the following statements about the taxonomic hierarchy is correct?

A. Category diversity decreases as we move from species to kingdom.

B. Category diversity increases as we move from species to kingdom. ✓

C. The number of individuals in a category decreases from species to kingdom.

D. All taxonomic categories contain an equal number of species.

Solution: Broader/higher categories (e.g., Kingdom) accommodate far greater organismal diversity, while Species is the most specific, homogeneous category.

Question 2

HARD

Identification and characterisation is comparatively EASIEST for which pair of taxonomic categories, owing to the high degree of shared characters?

A. Kingdom and Phylum

B. Genus and Species ✓

C. Class and Order

D. Phylum and Class

Solution: Lower categories like Genus and Species share far more characteristics among their members, making them easier to characterise accurately than broader categories.

Question 3

HARD

Match Column-I with Column-II and select the correct option.
Column-I: (a) Division (b) Class (c) Order (d) Family
Column-II: (i) Poaceae (ii) Insecta (iii) Angiospermae (iv) Primata

A. a-iii, b-ii, c-iv, d-i ✓

B. a-i, b-ii, c-iii, d-iv

C. a-ii, b-iii, c-i, d-iv

D. a-iv, b-i, c-ii, d-iii

Solution: Division = Angiospermae (used instead of Phylum for plants); Class = Insecta; Order = Primata; Family = Poaceae.

Question 4

HARD

Which observation argues AGAINST reproduction being an obligatory feature of every individual living organism?

A. Unicellular organisms reproduce by binary fission.

B. Mules, though alive, are sterile and cannot reproduce. ✓

C. Flowering plants reproduce sexually via seeds.

D. Bacteria multiply rapidly under favourable conditions.

Solution: Sterile individuals such as mules demonstrate that reproduction, while a characteristic of species/populations, is not true of every single living organism.

Question 5

HARD

Select the CORRECT statement distinguishing classical from modern taxonomy.

A. Classical taxonomy incorporates biochemical and physiological data extensively.

B. Modern taxonomy relies exclusively on gross morphological features.

C. Classical taxonomy is based mainly on outward, easily observable morphological features. ✓

D. Modern taxonomy discards evolutionary relationships entirely.

Solution: Classical (descriptive) taxonomy relied largely on gross external morphology, while modern taxonomy also integrates internal structure, cell structure, physiology, biochemistry and phylogeny.

Question 6

HARD

Match Column-I with Column-II and select the correct option.
Column-I: (a) Taxon (b) Systematics (c) Nomenclature (d) Identification
Column-II: (i) Assigning a correct scientific name (ii) A unit/rank of classification (iii) Determining the correct category of an unknown organism (iv) Study of diversity and evolutionary relationships

A. a-ii, b-iv, c-i, d-iii ✓

B. a-i, b-ii, c-iii, d-iv

C. a-iv, b-i, c-ii, d-iii

D. a-ii, b-i, c-iv, d-iii

Solution: Taxon = a rank/unit of classification; Systematics = study of diversity and evolutionary relationships; Nomenclature = assigning scientific names; Identification = determining correct category placement.

Question 7

HARD

Select the INCORRECT statement about Kingdom Protista.

A. Protists possess a well-defined nucleus and membrane-bound organelles.

B. Some protists use flagella or cilia for locomotion.

C. Protista is a clearly monophyletic group with no overlap with other kingdoms. ✓

D. Protista acts as a link connecting Monera, Fungi, Plantae and Animalia.

Solution: Protista is widely regarded as a 'kingdom of convenience' with poorly defined, overlapping boundaries — it is not a cleanly monophyletic group.

Question 8

HARD

Which of the following statements about Mycoplasma is INCORRECT?

A. They completely lack a cell wall.

B. They are the smallest known living cells.

C. They can survive without oxygen.

D. They possess a rigid peptidoglycan cell wall like typical bacteria. ✓

Solution: Mycoplasma are unique among prokaryotes for entirely lacking a cell wall — this is exactly why option D is incorrect.

Question 9

HARD

The rigid cell wall of most eubacteria (true bacteria) is primarily composed of:

A. Cellulose

B. Chitin

C. Peptidoglycan (murein) ✓

D. Pectin

Solution: Eubacterial cell walls are made of peptidoglycan (murein), a polymer distinct from the cellulose walls of plants or chitin walls of fungi.

Question 10

HARD

Woese's three-domain classification system separates Archaea from Bacteria primarily on the basis of:

A. Differences in ribosomal RNA (rRNA) sequences ✓

B. Differences in body size

C. Presence or absence of a cell wall

D. Habitat preference (terrestrial vs aquatic)

Solution: The three-domain system, based on comparative rRNA sequence analysis, recognises Archaea as fundamentally distinct from Bacteria, despite both being prokaryotic.

Question 11

HARD

Whittaker's five-kingdom classification placed Fungi in a separate kingdom mainly due to their:

A. Autotrophic mode of nutrition similar to plants

B. Absorptive (saprophytic) mode of nutrition and chitinous cell wall ✓

C. Complete absence of a cell wall

D. Prokaryotic cell organisation

Solution: Fungi were separated from Plantae in the five-kingdom system chiefly due to their heterotrophic, absorptive (saprophytic) nutrition and chitin-based cell walls.

Question 12

HARD

Assertion (A): Mycoplasma are often described as pathogenic in both plants and animals.
Reason (R): Despite lacking a cell wall, they can still cause infections and disease symptoms in their hosts.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are true; their wall-less nature does not prevent pathogenicity, and this fact appropriately supports the assertion about their disease-causing potential.

Question 13

HARD

A phylogenetic system of classification, as opposed to an artificial system, groups organisms primarily based on:

A. Superficial morphological similarity alone

B. Evolutionary relationships and common ancestry ✓

C. Economic importance to humans

D. Alphabetical order of scientific names

Solution: Phylogenetic classification reflects evolutionary relationships and shared ancestry between organisms, unlike artificial systems based on a few convenient morphological traits.

Question 14

HARD

Assertion (A): Families like Solanaceae and Convolvulaceae are grouped under the order Polymoniales.
Reason (R): An order is formed by aggregating families that share a few similar characters along with several dissimilar ones.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct — orders are formed by combining related families sharing some but not all characters, which correctly explains why Solanaceae and Convolvulaceae fall under Polymoniales.

Question 15

HARD

Which of the following statements about growth as a characteristic of life is INCORRECT?

A. Growth in living organisms is intrinsic (from within).

B. Growth in non-living objects such as crystals is also possible.

C. Intrinsic growth can occur without an increase in cell number, via an increase in cell mass/size.

D. Growth is possible only when new material is added from OUTSIDE the body, as in crystals or mountains. ✓

Solution: Growth in living organisms is intrinsic and can occur via an increase in cell number and/or cell size, unlike non-living objects, which grow only by external accumulation of material.

Phylum Protozoa

Question 16

HARD

Select the INCORRECT statement about Entamoeba histolytica.

A. It is an intestinal parasite of humans, causing amoebiasis (amoebic dysentery).

B. It moves and engulfs food using pseudopodia.

C. It possesses cilia over its entire body surface for locomotion. ✓

D. It belongs to the amoeboid group (Sarcodina) within Protozoa.

Solution: Entamoeba is an amoeboid protozoan that moves via pseudopodia, not cilia; ciliary locomotion is instead characteristic of ciliates like Paramecium.

Question 17

HARD

Which of the following best describes locomotion and feeding in Paramecium?

A. Uses cilia for locomotion and a cytostome (cell mouth) for feeding ✓

B. Uses pseudopodia for both locomotion and feeding

C. Uses flagella exclusively for locomotion and lacks any distinct feeding structure

D. Is completely non-motile and absorbs nutrients directly across its surface

Solution: Paramecium, a ciliated protozoan, moves using numerous cilia and takes in food through a specialised oral groove/cytostome.

Phylum Porifera

Question 18

HARD

In sponges, water enters the spongocoel through minute pores called ______ and exits through a larger opening called the ______.

A. Osculum; Ostia

B. Ostia; Osculum ✓

C. Ostia; Spongocoel

D. Spongocoel; Ostia

Solution: Water enters through numerous small incurrent pores (ostia) and is expelled through a single large opening, the osculum.

Question 19

HARD

The spongocoel (central cavity) of sponges is lined by which type of cell, responsible for generating water currents?

A. Choanocytes (collar cells) ✓

B. Pinacocytes

C. Amoebocytes

D. Archaeocytes

Solution: Choanocytes, flagellated collar cells lining the spongocoel and canals, create water currents used in filter feeding and gas exchange.

Question 20

HARD

Which of the following statements about phylum Porifera is INCORRECT?

A. They exhibit a cellular level of organisation without true tissues.

B. Sexes are usually separate (dioecious). ✓

C. The body is perforated by numerous minute pores.

D. Development includes a free-swimming larval stage.

Solution: Sponges are generally hermaphrodites (monoecious/bisexual), not typically dioecious with separate sexes — making this the incorrect statement.

Coelenterata & Ctenophora

Question 21

HARD

Which coelenterate class possesses an EXOSKELETON of calcium carbonate, forming the structural basis of coral reefs?

A. Hydrozoa

B. Scyphozoa

C. Anthozoa ✓

D. Ctenophora

Solution: Reef-building (hermatypic) corals belong to class Anthozoa, secreting a hard calcium carbonate exoskeleton that builds up as coral reefs over time.

Question 22

HARD

Which of the following is a feature shared by BOTH Coelenterata and Ctenophora?

A. Presence of nematocysts (stinging cells)

B. Radial/biradial symmetry combined with tissue-level (diploblastic) organisation ✓

C. Presence of a true coelom

D. Exclusively freshwater habitat

Solution: Both phyla are diploblastic with radial (Coelenterata) or biradial (Ctenophora) symmetry. Nematocysts are unique to Coelenterata (ctenophores instead have colloblasts); neither has a true coelom, and Ctenophora is exclusively marine (Coelenterata is mostly marine with a few freshwater exceptions like Hydra).

Phylum Platyhelminthes

Question 23

HARD

Which of the following statements about Planaria is INCORRECT?

A. It shows a remarkable capacity for regeneration.

B. It is a free-living flatworm, unlike most other Platyhelminthes.

C. It possesses a well-developed true coelom for organ support. ✓

D. It shows bilateral symmetry and cephalisation.

Solution: Like all Platyhelminthes, Planaria is acoelomate — it lacks a true coelom, despite being free-living rather than parasitic.

Question 24

HARD

Which of the following correctly explains why Platyhelminthes are termed 'acoelomate'?

A. They completely lack a mesoderm layer.

B. The space between body wall and gut is filled with parenchymatous cells rather than a fluid-filled cavity. ✓

C. Their body cavity is lined by mesoderm only on the outer (body-wall) side.

D. Their coelom develops through the enterocoelous method.

Solution: Acoelomates like flatworms ARE triploblastic (mesoderm is present) but lack any body cavity between the body wall and gut — the space is instead packed with mesoderm-derived parenchyma.

Question 25

HARD

Which of the following phyla is triploblastic but ACOELOMATE?

A. Annelida

B. Aschelminthes

C. Platyhelminthes ✓

D. Mollusca

Solution: Platyhelminthes (flatworms) are triploblastic but lack any body cavity, making them acoelomate — unlike Annelida and Mollusca (true coelomate) or Aschelminthes (pseudocoelomate).

Phylum Aschelminthes

Question 26

HARD

Which of the following statements about Wuchereria bancrofti (filarial worm) is CORRECT?

A. It causes elephantiasis by chronically inflaming the lymphatic vessels, especially of the limbs. ✓

B. It is a free-living, non-parasitic roundworm.

C. It is transmitted through contaminated drinking water.

D. It belongs to phylum Platyhelminthes.

Solution: Wuchereria, transmitted by Culex mosquitoes, causes lymphatic filariasis (elephantiasis) through slow, chronic inflammation of lymph vessels, most visibly in the limbs.

Question 27

HARD

Select the INCORRECT statement about Ascaris lumbricoides (roundworm).

A. It is a common intestinal parasite of humans.

B. Sexes are separate, with males usually smaller than females.

C. It possesses flame cells for excretion, similar to flatworms. ✓

D. Its body cavity is a pseudocoelom.

Solution: Flame cells are a hallmark of Platyhelminthes; Aschelminthes such as Ascaris have a different excretory arrangement and lack flame cells.

Phylum Annelida

Question 28

HARD

Which of the following statements correctly compares segmentation in Nereis and Hirudinaria (leech)?

A. In Nereis, external segments correspond to internal septa; in Hirudinaria, external annulations are more numerous than the true internal segments. ✓

B. Nereis shows no segmentation at all, while Hirudinaria is fully segmented.

C. Both show an identical correspondence between external and internal segmentation.

D. Neither Nereis nor Hirudinaria shows any form of segmentation.

Solution: In Nereis, each external ring corresponds to an internal septum (true metamerism). In Hirudinaria, the external surface bears more numerous ring-like annulations than the actual number of internal segments — the correspondence is inexact.

Question 29

HARD

Which of the following animals possesses a CLOSED type of circulatory system, with blood confined to definite vessels?

A. Pheretima (earthworm) ✓

B. Palaemon (prawn)

C. Periplaneta (cockroach)

D. Pila (snail)

Solution: Annelids such as the earthworm have a closed circulatory system. Arthropods (prawn, cockroach) and most molluscs (snail) instead show an open circulatory system with a haemocoel.

Phylum Arthropoda

Question 30

HARD

Mandibles (jaws) used for biting and chewing food are present in which of the following arthropod groups?

A. Insecta and Crustacea ✓

B. Arachnida only

C. Merostomata only

D. All arthropods lack jaws

Solution: Insects and crustaceans possess mandibles for feeding; arachnids (spiders, scorpions) and Merostomata (king crab) instead bear chelicerae, not true mandibulate jaws.

Question 31

HARD

Which of the following pairs is CORRECTLY matched?

A. Palaemon – Prawn ✓

B. Bombyx – House fly

C. Limulus – Housefly killer

D. Laccifer – Silk moth

Solution: Palaemon is the prawn. Bombyx is the silkworm moth (not housefly); Limulus is the king crab; Laccifer is the lac insect (not silk moth).

Question 32

HARD

Three of the following four genera belong to class Insecta; select the ONE that does NOT.

A. Apis

B. Bombyx

C. Limulus ✓

D. Laccifer

Solution: Apis (honeybee), Bombyx (silk moth) and Laccifer (lac insect) are all insects; Limulus (king crab) belongs to class Merostomata, not Insecta.

Question 33

HARD

Assertion (A): Insects (class Insecta) show greater species diversity than any other class within Arthropoda.
Reason (R): Insects have successfully colonised terrestrial, aquatic and aerial habitats, aided by adaptations like wings and a hardened exoskeleton.

- **A. Both A and R are true and R is the correct explanation of A. ✓**
- **B.** Both A and R are true but R is NOT the correct explanation of A.
- **C.** A is true but R is false.
- **D.** A is false but R is true.

Solution: Both statements are true; the arthropod feature of a versatile, modifiable body plan (wings, hardened cuticle) correctly explains why insects have radiated into such enormous diversity.

Phylum Mollusca

Question 34

HARD

In molluscs, the muscular ventral structure primarily used for locomotion is called the:

- **A.** Mantle
- **B. Foot ✓**
- **C.** Radula
- **D.** Osphradium

Solution: The muscular foot is the primary locomotory organ in molluscs, variously modified for crawling, burrowing or swimming across different classes.

Question 35

HARD

Match the genus with its common name and select the correct option.
(i) Chiton (a) Tusk shell
(ii) Dentalium (b) Coat-of-mail shell
(iii) Aplysia (c) Sea hare
(iv) Sepia (d) Cuttlefish

A. i-b, ii-a, iii-c, iv-d ✓

B. i-a, ii-b, iii-d, iv-c

C. i-c, ii-d, iii-a, iv-b

D. i-d, ii-c, iii-b, iv-a

Solution: Chiton = coat-of-mail shell, Dentalium = tusk shell, Aplysia = sea hare, Sepia = cuttlefish → i-b, ii-a, iii-c, iv-d.

Question 36

HARD

Which of the following pairs is CORRECTLY matched?

A. Octopus – Devilfish ✓

B. Pila – Cuttlefish

C. Loligo – Apple snail

D. Sepia – Squid

Solution: Octopus is commonly called devilfish. Pila is the apple snail (not cuttlefish); Loligo is the squid (not apple snail); Sepia is the cuttlefish (not squid).

Phylum Echinodermata

Question 37

HARD

Antedon (sea lily) differs from Asterias (starfish) most notably in its:

- **A. Sessile or free-swimming, stalked/feathery body form, placed in class Crinoidea ✓**
- **B.** Complete absence of a water vascular system
- **C.** Radially asymmetric body plan, unlike other echinoderms
- **D.** Segmented body plan resembling annelids

Solution: Antedon belongs to class Crinoidea, which characteristically shows a stalked or free-swimming, feathery body form quite distinct from the star-shaped, free-moving Asterias (Asteroidea).

Question 38

HARD

Which of the following is NOT a characteristic feature of phylum Echinodermata?

- **A.** Spiny, calcareous endoskeleton
- **B.** Water vascular system derived from the coelom
- **C. A well-developed, distinct excretory system ✓**
- **D.** Exclusively marine habitat

Solution: Echinoderms notably LACK a distinct excretory system; nitrogenous waste is removed largely by diffusion across the body surface.

Phylum Hemichordata

Question 39

HARD

Balanoglossus, a member of phylum Hemichordata, has a body divided into which three regions?

- **A.** Head, thorax, abdomen
- **B. Proboscis, collar, trunk ✓**
- **C.** Cephalothorax, abdomen, tail
- **D.** Anterior, middle and posterior segments only

Solution: The hemichordate body plan is divided into three distinct regions: an anterior proboscis, a collar, and a long trunk.

Chordata – General Characters

Question 40

HARD

Which of the following statements correctly compares notochord retention across chordate subphyla?

A. Urochordata retains the notochord throughout life in the adult trunk region.

B. Cephalochordata retains the notochord throughout life, extending from head to tail. ✓

C. Vertebrata retains an unmodified notochord throughout adult life in all members.

D. None of the subphyla retain the notochord, even in larval stages.

Solution: In Urochordata the notochord is present only in the larval tail and is lost in adults; in Cephalochordata it persists lifelong; in Vertebrata it is typically replaced by a vertebral column during development.

Question 41

HARD

In which of the following groups is the embryonic notochord typically replaced by a bony or cartilaginous vertebral column?

A. Urochordata

B. Cephalochordata

C. Vertebrata ✓

D. Hemichordata

Solution: In Vertebrata, the notochord present in the embryo is replaced by a vertebral column (cartilaginous or bony) as development proceeds.

Protochordata

Question 42

HARD

Which of the following chordates LACK both a notochord and a vertebral column in their adult stage?

- **A. Members of Urochordata (e.g., Herdmania) ✓**
- **B.** Members of Cephalochordata (e.g., Branchiostoma)
- **C.** Members of Vertebrata (e.g., Rana)
- **D.** All chordates retain a notochord in adulthood

Solution: In adult urochordates (tunicates), the larval notochord is lost and no vertebral column develops — unlike Cephalochordata (notochord persists) or Vertebrata (notochord replaced by a vertebral column).

Question 43

HARD

Herdmania and Salpa are examples of which subphylum of Protochordata?

- **A.** Cephalochordata
- **B. Urochordata (Tunicata) ✓**
- **C.** Vertebrata
- **D.** Hemichordata

Solution: Herdmania and Salpa are both tunicates, placed under subphylum Urochordata.

Vertebrata – Cyclostomata

Question 44

HARD

The skin of cyclostomes (e.g., Petromyzon) is typically:

- **A.** Covered with placoid scales
- **B.** Covered with bony scales like Osteichthyes
- **C. Smooth and devoid of scales, but rich in mucous glands ✓**
- **D.** Covered with feathers

Solution: Cyclostomes have smooth, scaleless skin with numerous mucous glands, unlike both cartilaginous fishes (placoid scales) and bony fishes (bony scales).

Question 45

HARD

Cyclostomes such as Petromyzon and Myxine are typically ectoparasites primarily on:

A. Terrestrial mammals

B. Certain species of fishes ✓

C. Amphibians only

D. Birds

Solution: Marine cyclostomes such as lampreys and hagfish attach to and feed on certain fish species as ectoparasites, using their sucker-like mouths.

Superclass Pisces

Question 46

HARD

Which of the following is an exclusively marine bony fish?

A. Labeo

B. Catla

C. Exocoetus (flying fish) ✓

D. Clarias

Solution: Exocoetus, the flying fish, is a marine bony fish; Labeo, Catla and Clarias are all freshwater fishes.

Question 47

HARD

A fish possessing a cartilaginous endoskeleton, placoid scales, and lacking both an operculum and an air bladder most likely belongs to which group?

A. Osteichthyes

B. Chondrichthyes ✓

C. Cyclostomata

D. Amphibia

Solution: This combination of features — cartilage skeleton, placoid scales, no operculum, no air bladder — is diagnostic of class Chondrichthyes (cartilaginous fishes).

Question 48

HARD

From the following list, count the number of BONY fishes (Osteichthyes): Hippocampus, Pristis, Anguilla, Torpedo, Labeo, Scoliodon, Exocoetus, Carcharodon

- **A.** Three
- **B. Four ✓**
- **C.** Five
- **D.** Six

Solution: Bony fishes here are Hippocampus, Anguilla, Labeo and Exocoetus (4 total). Pristis, Torpedo, Scoliodon and Carcharodon are cartilaginous fishes (Chondrichthyes).

Question 49

HARD

Which of the following pairs of fish BOTH possess an operculum and an air bladder?

- **A. Labeo and Exocoetus ✓**
- **B.** Scoliodon and Pristis
- **C.** Torpedo and Carcharodon
- **D.** Trygon and Scoliodon

Solution: Labeo and Exocoetus are bony fishes (Osteichthyes), possessing both an operculum and an air bladder; the fish in the other options are cartilaginous (Chondrichthyes), which lack these structures.

Question 50

HARD

Which of the following groups includes species found in BOTH freshwater and marine habitats?

- **A. Class Pisces (e.g., Labeo in freshwater, Exocoetus in marine water) ✓**
- **B.** Phylum Ctenophora (exclusively marine)
- **C.** Phylum Echinodermata (exclusively marine)
- **D.** Class Cyclostomata (exclusively marine)

Solution: Class Pisces spans both freshwater genera (e.g., Labeo, Catla) and marine genera (e.g., Exocoetus, Scoliodon), unlike the other listed groups, which are exclusively marine.

Question 51

HARD

Which of the following sets consists of animals that are ALL exclusively marine?

A. Physalia, Aurelia, Pristis, Antedon ✓

B. Hydra, Physalia, Labeo, Antedon

C. Physalia, Aurelia, Ascaris, Antedon

D. Hydra, Ascaris, Pila, Antedon

Solution: Physalia (Portuguese man-of-war), Aurelia (jellyfish), Pristis (saw fish) and Antedon (sea lily) are all exclusively marine. Hydra is freshwater; Labeo is a freshwater fish; Ascaris and Pila are not marine.

Class Amphibia

Question 52

HARD

Which of the following statements about respiration in Amphibia is CORRECT?

A. Respiration occurs exclusively through lungs at all life stages.

B. Tadpole larvae respire through gills; adults respire through lungs, skin and buccal lining. ✓

C. Adults respire exclusively through gills, similar to fish.

D. Amphibians completely lack any form of cutaneous respiration.

Solution: Amphibian respiration changes across the life cycle — aquatic larvae (tadpoles) use gills, while terrestrial adults use lungs, supplemented by the moist skin and buccal cavity lining.

Question 53

HARD

Which of the following features is common to BOTH class Amphibia and class Reptilia?

A. A three-chambered heart (in most members) ✓

B. Cornified, scaly, dry skin

C. Exclusively aquatic development of eggs

D. Presence of an amniotic egg

Solution: Most amphibians and most reptiles (except crocodiles) share a three-chambered heart. Dry scaly skin and amniotic eggs are reptilian features not shared by amphibians.

Question 54

HARD

Salamandra, an amphibian that retains its tail throughout life, belongs to which order?

A. Anura

B. Urodela (Caudata) ✓

C. Apoda

D. Chelonia

Solution: Salamandra belongs to order Urodela (Caudata), the tailed amphibians, distinct from the tailless Anura and limbless Apoda.

Question 55

HARD

Select the correct sequence of taxonomic categories (Kingdom to Species) for Rana tigrina (frog).

A. Animalia → Chordata → Amphibia → Anura → Ranidae → Rana → tigrina ✓

B. Animalia → Chordata → Anura → Amphibia → Ranidae → Rana → tigrina

C. Animalia → Amphibia → Chordata → Anura → Ranidae → Rana → tigrina

D. Chordata → Animalia → Amphibia → Anura → Ranidae → Rana → tigrina

Solution: Correct hierarchy: Kingdom-Animalia, Phylum-Chordata, Class-Amphibia, Order-Anura, Family-Ranidae, Genus-Rana, Species-tigrina.

Class Reptilia

Question 56

HARD

Which of the following statements about class Reptilia is CORRECT?

A. Most reptiles are oviparous, though some (e.g., certain lizards and snakes) are ovoviviparous. ✓

B. All reptiles possess a four-chambered heart.

C. Reptiles have moist, glandular skin similar to amphibians.

D. Fertilization in reptiles is always external.

Solution: Most reptiles lay eggs (oviparous), but some lizards and snakes retain eggs internally until hatching (ovoviviparous). Only crocodiles have a near-complete four-chambered heart, skin is dry and scaly, and fertilization is internal.

Question 57

HARD

Poison (venom) glands, used for both offence and defence, are present in which reptilian group?

A. Turtles (Chelonia)

B. Certain snakes (e.g., cobra, viper) ✓

C. Crocodiles

D. All lizards without exception

Solution: Venom glands, modified salivary glands, are found in certain venomous snakes such as cobras and vipers, used both to subdue prey and for defence.

Question 58

HARD

Which of the following statements about Chelone (turtle) is CORRECT?

A. It possesses a bony shell fused with the vertebral column and ribs. ✓

B. It has a soft, unprotected body without any shell.

C. It lacks a cornified skin covering.

D. It is exclusively found in freshwater, with no marine representatives.

Solution: Turtles possess a bony carapace (dorsal) and plastron (ventral) fused with the vertebral column and ribs; marine turtles also exist, so option D is also incorrect.

Class Aves

Question 59

HARD

In birds, epidermal scales (a reminder of their reptilian ancestry) are typically found on the:

A. Wings and back

B. Legs (tarsometatarsus) and toes ✓

C. Beak surface only

D. Entire body, similar to reptiles

Solution: Birds retain epidermal scales on their legs (tarsometatarsus) and toes, a vestige of their reptilian lineage, while the rest of the body is covered in feathers.

Question 60

HARD

Select the set in which ALL the listed animals are NON-homeothermic (poikilothermic/cold-blooded).

A. Columba, Corvus, Pteropus

B. Calotes, Chelone, Labeo ✓

C. Macropus, Delphinus, Equus

D. Struthio, Corvus, Psittacula

Solution: Calotes (lizard), Chelone (turtle) and Labeo (fish) are all poikilothermic. The other options include birds (Columba, Corvus, Struthio, Psittacula) or mammals (Pteropus, Macropus, Delphinus, Equus), all of which are homeothermic.

Question 61

HARD

Which of the following statements about reproduction in class Aves is CORRECT?

A. All birds are viviparous.

B. Birds are exclusively oviparous, laying hard-shelled, calcareous eggs. ✓

C. Fertilization in birds is always external.

D. Birds show no parental care of eggs or young.

Solution: Birds are exclusively oviparous, laying eggs with a protective, hard calcareous shell; fertilization is internal, and most birds show extensive parental care.

Class Mammalia

Question 62

HARD

Select the set in which ALL the animals show a viviparous mode of reproduction.

A. Ornithorhynchus, Macropus, Equus

B. Macropus, Equus, Delphinus ✓

C. Tachyglossus, Ornithorhynchus, Chelone

D. Corvus, Macropus, Equus

Solution: Macropus (kangaroo), Equus (horse) and Delphinus (dolphin) are all viviparous mammals. Ornithorhynchus and Tachyglossus are oviparous monotremes; Chelone and Corvus are oviparous non-mammals.

Question 63

HARD

Which of the following represents a CORRECT combination of exclusively aquatic mammals?

- **A. Delphinus, Physeter, Orcinus ✓**
- **B.** Pteropus, Loris, Physeter
- **C.** Camelus, Physeter, Macropus
- **D.** Vespertilio, Orcinus, Equus

Solution: Delphinus (dolphin), Physeter (sperm whale) and Orcinus (killer whale) are all exclusively aquatic mammals. The other options mix in bats, primates, camels, kangaroos or horses.

Question 64

HARD

Which of the following animals is MONOECIOUS (hermaphroditic), possessing both male and female reproductive organs in the same individual?

- **A. Pheretima (earthworm) ✓**
- **B.** Periplaneta (cockroach)
- **C.** Rana (frog)
- **D.** Columba (pigeon)

Solution: Earthworm (Pheretima) is a classic hermaphrodite, along with animals like leech and tapeworm. Cockroach, frog and pigeon are all dioecious (sexes separate).

Question 65

HARD

From the following list, count the number of POIKILOTHERMIC (cold-blooded) animals: Calotes, Columba, Labeo, Equus, Chelone, Corvus, Rana, Delphinus

- **A.** Three
- **B. Four ✓**
- **C.** Five
- **D.** Six

Solution: Poikilothermic animals here: Calotes (lizard), Labeo (fish), Chelone (turtle), Rana (frog) = 4. Columba and Corvus are birds; Equus and Delphinus are mammals — all homeothermic.
